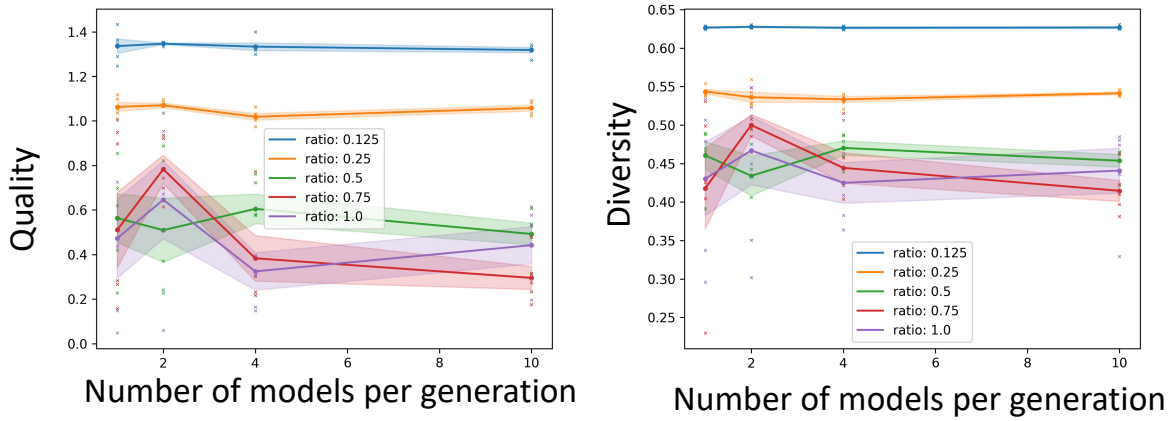

Figure 9: **The effect of synthetic data ratio on shifts in diversity and quality holds when increasing the number of models per generation.**

C.2 The effect of manipulating dataset quality on the distribution shift dynamics

In this experiment, we explore the hypothesis that data quality is one of the potential factors influencing distribution shifts. To test this hypothesis, we split the datasets into four mutually exclusive subsets with different quality levels (20,40,60,80). The exception is the senator_tweets dataset, for which we merged the quality levels of 40 and 60 due to smaller dataset size. Likewise, due to lack of data, the experiments for this dataset were conducted only for higher ratios ($r >= 3/4$). Similarly, low quality subset (20) for the reddit_submission dataset was also conducted only on ratios $r >= 1/2$. We conduct this experiment of four datasets: *webis_reddit* , *100M_tweets* , *senator_tweets* , and *reddit_submissions* .

C.2.1 Effect on quality

Figure 10 shows the quality values of iterative chains for different quality levels. As in section 4.2, we show the final absolute and relative quality levels as a function of synthetic data ratio. Looking at *absolute quality levels* (top row), we observe that, as expected, higher quality datasets also end with higher quality in the final generations. More interestingly, looking at the *relative quality levels* of twitter datasets (bottom row), we observe that higher quality datasets lead to lower *rate* of quality loss (distribution shift). That is, not only does higher input data quality increase the quality of the final generated dataset, it also decreases the percentage of original quality lost due to recursive training. Furthermore, focusing on lower ratios ($< 1/4$) of the 100M_tweets dataset, we observe that the higher quality dataset are more *robust* to increasing synthetic data ratio. For the quality of 80, major quality losses are observed only at $r = 1/4$, while for lower qualities it is observed already at $r = 1/8$. Finally, it should also be noted that the 100M_tweets dataset with quality 20 does not appear to lead to significant shifts with higher synthetic data ratios. Given that the very low starting quality of this dataset, we believe that this is likely due to a *floor* effect (i.e. there are no losses in quality because the initial dataset was already close to the lower bound. Curiously, on both Reddit datasets, we do not observe strong differences when manipulating quality, implying that there are other factors than quality influencing collapse. A similar study focusing on losses of diversity is presented in Appendix C.2.2, where an effect is observed only on the *senator_tweets* dataset. Overall, this experiment shows that, in some conditions, high-quality datasets lead to increased robustness to distribution shifts.

C.2.2 Effect on diversity

Figure 11 shows the semantic diversity values of iterative chains for different quality levels. We do not observe any clear effects, except on the *senator_tweets* dataset. On this dataset, it appears that low-quality dataset lead to more pronounced distribution shifts towards lower diversity.

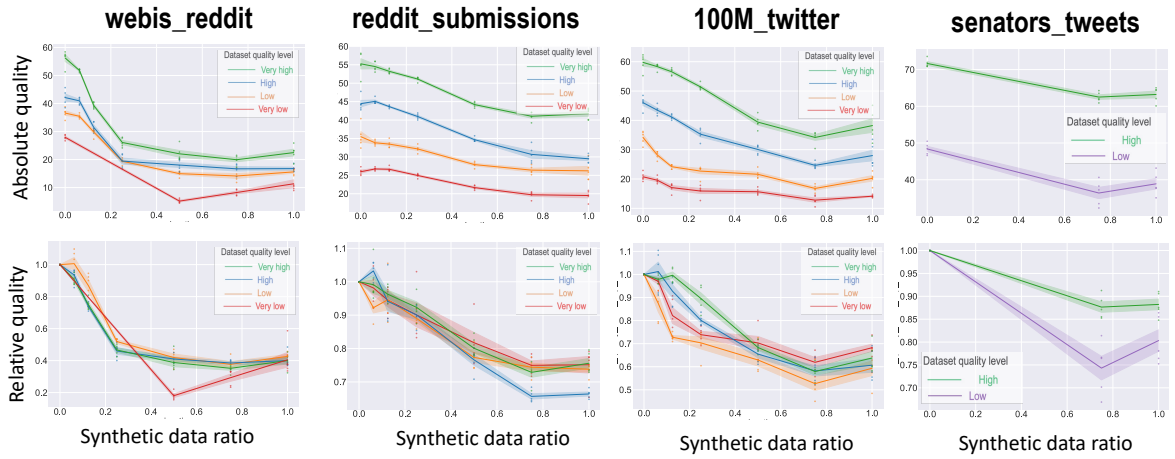

Figure 10: **Effect of human data quality on the *rate* of degradation and sensitivity to synthetic data** Absolute measures (top row) correspond to the value of the corresponding metric at generation 19. Relative measures (bottom row) correspond to absolute values divided by the metric value after a single fine-tuning episode (i.e. generation 0). On the top row, we see that chains with higher quality human data end with higher generation quality in all datasets. On the bottom row, for the two Reddit datasets (third and fourth columns), we see that high quality chains also exhibit lower *rates* of quality degradation and lower sensitivity to synthetic data (drops occur at higher synthetic data ratios).

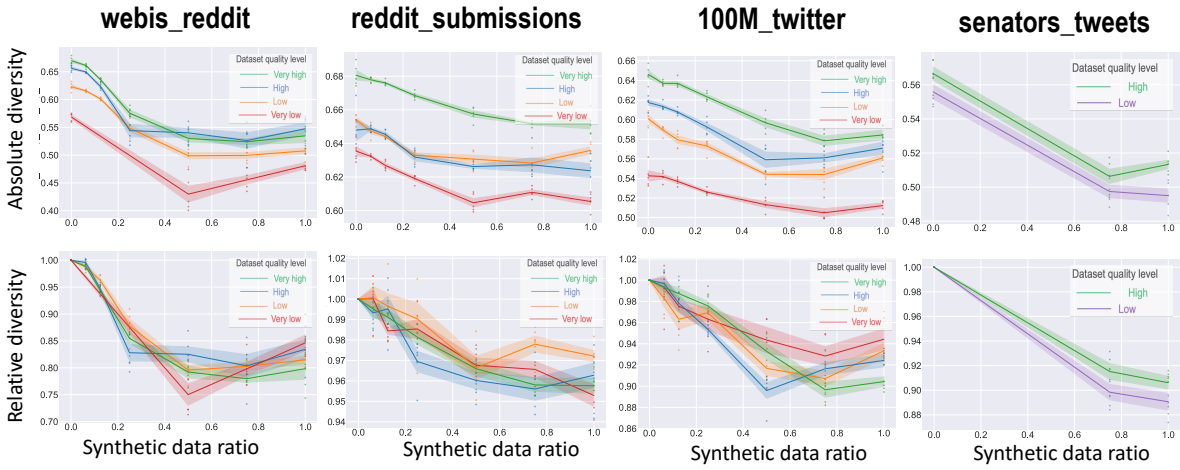

Figure 11: **Effect of manipulating dataset quality on sensitivity to synthetic data ratio, for four different datasets.** Absolute measures correspond to the value of the corresponding metric at generation 19. Relative measure correspond to absolute values divided by the metric value after a single fine-tuning episode (i.e. generation 0).No clear effect is observed, except potentially on the *senator_tweets* dataset.

C.3 Toy model exploring the causes of the non-linear relationship between diversity loss and synthetic-data ratio

In section 4.2, we observed that for the Wikipedia dataset, the relationship between diversity loss and synthetic-data ratio was non-linear. Indeed, the greatest drops in diversity are observed for intermediate synthetic-data ratios, rather than for high values as in other datasets and previous works (Bertrand et al., 2023; Bohacek and Farid, 2023; Kazdan et al., 2024). While at first surprising, we believe this pattern can be explained if we assume that synthetic data aligns more with the models' priors. This is not a strong assumption given that that data was generated by other fine-tuned versions of the same base models. The intuition, which we experimentally confirm below, is that those datapoints aligned with the model's priors, have a stronger effect on the training process. And this then leads to the intermediate synthetic data ratios to essen-

tially learn from less data. Let us consider the three different synthetic-data ratios:

- When synthetic-data ratio is low: While the model preferentially learns from synthetic data, most of its training data is human generated. Therefore, the model learns a distribution that resembles the human data despite this bias.
- When synthetic-data ratio is high: the model preferentially learns from synthetic data, but anyway most of the training data is synthetic. This bias thus does not have a large effect.
- When synthetic-data ratio is intermediate: the model preferentially learns from synthetic data, and receives human data and synthetic in comparable proportion. However, the bias will lead the model to essentially discard human data, and to learn only from the synthetic data, just like for high synthetic-data ratios. The difference is that the pool of synthetic data to learn from is here lower than in the high synthetic-data ratio.

To test this hypothesis, we develop a toy example where we could manipulate whether learning from synthetic data aligned with models' priors is favored. In this model, the true (*human*) distribution is a uniform distribution over integers in $[0, N]$. The *model* is implemented as a normalized histogram over training datapoints, which are a combination of true and synthetic datapoints from the previous generations. We sample N points from the true distribution, and normalize the resulting histogram to get the first model. Then, we sample $r * N$ points from this model, and $(1 - r) * N$ points from the true distribution. We again derive a probability distribution from this sample to get the new model. We repeat this process for 20 time steps.

To introduce the bias mentioned in our hypothesis, we assume that the models have a prior to sample multiples of 2. We thus multiply the histogram by a corresponding bias vector before normalizing. Additionally, we can manipulate whether the human data overlaps with this bias: we can modify the true distribution so that it does not contain multiples of 2.

We then ran 50 simulations for different values of synthetic-data ratio, manipulating whether learning is biased and whether the true distribution overlaps with this bias. As show in Figure 12, we observe that when learning is biased and the true distribution does not overlap (right column), we can reproduce the U-shape found for the Wikipedia dataset. This non-linear relationship disappears when we remove this bias (left column) or when the true distribution overlaps with this bias. We were able to observe this pattern both in the Accumulation (top-row) and no-Accumulation (bottom row) settings. This is therefore consistent with our hypothesis for explaining the observed U-shape relationship.

C.4 Regression models' explained variances

Here we provides r^2 measures for the linear regression models used in experiments in sections 4.3 and 4.4. Table 3 corresponds to table 1 from section 4.3. Table 4 corresponds to Figure 4 from section 4.4. We can see that the considered properties can partially account for observed effect, many more effects remain to be uncovered (in particular in the 100M tweets and wikipedia datasets).

C.5 Additional experiments on the distribution shift of political lean

We provide here the additional figures that are discussed in section 4.5. On Figure 13a we observe a steady increase in the proportion of "perfectly neutral" tweets (with assigned a score of exactly 50). On Figure 13b while we observe a slight increase in the number of non-political tweets, those remain marginal, indicating that the models are able to maintain the focus on political topics (Figure 13a). This suggests that generated tweets remain in the topic of politics, but drift from strong partisanship.

Figure 13a revealed that the shift is partly driven by an increase in politically neutral content. To isolate the different mechanisms at play, we performed the same analysis, but without taking politically neutral tweets into account (Figure 14). The consequence of this manipulation was to accentuate the observed asymmetry, as the political lean that minimizes shift magnitude moves even more toward the left. This suggests that there might two interacting mechanisms influencing political lean evolution: first, a tendency to generate politically-neutral content; and a tendency to shift the distribution toward left-wing content.

Experiments in the paper were conducted with mixed-model iterative chains - each generation a fresh base model is sampled out of four possible models options (`LLama-3.2-1B`, `Qwen2.5-1.5B`, `SmolLM-1.7B`, `Falcon3-1B-Base`). As different